



Pharmacy Inventory and Prescription Management System

Technical Documentation

**CSC316: Advanced Database
Systems**

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GROUP 10

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TABLE OF CONTENTS

SECTION 1: PROJECT OVERVIEW

SECTION 2: SYSTEM ARCHITECTURE

SECTION 3: DATABASE DESIGN

SECTION 4: GUI & FEATURES

SECTION 5: ADVANCED FEATURES

SECTION 6: INSTALLATION & SETUP

SECTION 7: PROJECT STATISTICS

SECTION 8: TROUBLESHOOTING

SECTION 1: PROJECT OVERVIEW

MedLife is a desktop-based pharmacy chain management system designed to streamline the operations of a multi-branch network. Built for scale and reliability, it addresses the challenges of fragmented data across different locations.

Key Features:

- **Medicine Registry:** Comprehensive cataloging of drugs including generic names, strengths, and controlled substance flags.
- **Inventory Management:** Track stock across all branches with automated low-stock and near-expiry alerts.
- **Prescription Tracking:** Link patients to doctors and track medication dispensing securely.
- **Multi-Branch Support:** Scaled to support 9 pharmacies across 6 different cities.
- **Analytics Dashboard:** Visualize stock levels, revenue trends, and doctor performance via interactive charts.
- **Advanced MongoDB Queries:** Extensive use of the aggregation framework for complex reporting.
- **Database Optimization:** Strategic use of indexes and query analysis tools.
- **Backup & Restore:** Native mongodump/mongorestore integration directly from the GUI.
- **ACID Transactions:** Multi-document atomic transactions ensuring reliable prescription dispensing.

SECTION 2: SYSTEM ARCHITECTURE

Project Structure:

Pharmacy Management System/
main.py

database/
 connection.py
 crud.py
 aggregations.py
 indexes.py
 transactions.py
 backup_restore.py
 setup_database.py

gui/
 dashboard.py
 medicines.py
 inventory.py
 generic.py
 settings.py

docs/
 README.md
 Requirements.txt
 SETUP_GUIDE.md

Technology Stack

Component	Technology / Tool
Language	Python 3.8+
GUI Framework	Tkinter
Database	MongoDB (NoSQL)
Database Driver	PyMongo
Data Visualization	Matplotlib

SECTION 3: DATABASE DESIGN

The database consists of 8 main collections, structured for fast reads and atomic updates.

Medicines

Schema: name, genericName, manufacturer, type, unitPrice, minimumStockThreshold, strength, requiresPrescription, controlledSubstance

Pharmacies

Schema: name, address, city, phone, email, licenseNumber, manager, established

Prescriptions

Schema: patientId, doctorId, medicineId, pharmacyId, prescriptionDate, quantityPrescribed, dosage, duration, status, dispensedAt

Inventory

Schema: medicineId, pharmacyId, currentStock, minimumThreshold, batches [batchNumber, manufacturingDate, expiryDate, quantityInBatch], lastRestocked

Persons

Schema: name, roles, specialization, contactNumber, email, address, city, licenseNumber

Supplyorders

Schema: supplierId, pharmacyId, medicineId, quantityOrdered, unitCost, totalCost, orderDate, deliveryDate, status, deliveryAddress

Suppliers

Schema: companyName, contactPerson, phone, email, address, rating, paymentTerms

Categories

Schema: name, description

SECTION 4: GUI & FEATURES

Dashboard Statistics

The Dashboard highlights real-time system metrics: 506 Medicines, 9 Pharmacies, 3000 Prescriptions, 150 Inventory records, 86 Low Stock alerts, and 84 Near Expiry alerts.

Visualizations

The Stock Level by Branch bar chart displays total and low stock metrics across locations such as Saddar, Saidpur Road, Gulberg, G-13, Bahria Town, and Iqbal Town.

Active Alerts

Low Stock list: Augmentin 625mg (30/50), Metformin 500mg (20/200), Insulin Glargine (10/30), Atorvastatin 20mg (90/100), Clonazepam 0.5mg (15/40), Salbutamol Inhaler (12/30), Sertraline 50mg (45/60), Clopidogrel 75mg (60/80).

Near Expiry list (90 days): Augmentin 625mg (01 Aug 2026), Metformin 500mg (30 May 2025), Insulin Glargine (10 Jun 2025), Salbutamol Inhaler (30 Jul 2025), Omeprazole 20mg (01 Jun 2025).

Navigation Sidebar

- **Overview:** Dashboard
- **Management:** Medicines, Pharmacies, Persons, Inventory, Prescriptions, Supply Orders, Suppliers
- **Analytics:** Analytics, Query Lab
- **Database:** Indexes, Optimization, Transactions, Backup/Restore

SECTION 5: ADVANCED FEATURES

(a) MongoDB Aggregation Pipelines

Features 12+ pre-built pipelines including Top Prescribed Medicines, Prescriptions per Doctor, Stock Level per Branch, Supplier Performance Ranking, Controlled Substance Flags, and Revenue per Pharmacy. The Analytics page displays Doctor Stats via bar charts (Dr. Ayesha Tariq, Dr. Zubair Qureshi, etc., each with ~500 prescriptions) and pie charts by specialization (General Physician 33%, Pulmonologist 17%, etc.).

(b) Index Management

Supports Single-field, Compound, Multikey, Text, TTL, and Expiry Date indexes. The GUI permits quick creation of Text Indexes (Medicines), Compound Indexes (Inventory), Multikey Indexes (Person Roles), and TTL Indexes (Supply Orders). Existing indexes on the medicines collection include `_id`, `name_1`, `genericName_1`, and `type_1`.

(c) ACID Transactions

A Transaction-Safe Prescription Dispense page manages concurrent dispensing operations. Using MongoDB's multi-document transaction API (`client.start_session()`, `session.start_transaction()`), the system validates stock, decrements inventory, and marks prescriptions as 'Dispensed' atomically. Requires a Replica Set setup for full ACID compliance.

(d) Backup & Restore

Integrated native backup workflows using `mongodump` and `mongorestore` processes, executable directly from the user interface.

(e) Query Lab

A custom aggregation pipeline JSON editor with a collection selector, allowing administrators to execute arbitrary read-only queries.

SECTION 6: INSTALLATION & SETUP

Prerequisites

- Python 3.8+
- MongoDB 4.4+ (running on localhost:27017)

Installation Steps

Requirement	Specification
Operating System	Windows 10+ / macOS 10.14+ / Linux
Python Version	3.8 or higher
Database	MongoDB 4.4 or higher
RAM	2GB Minimum, 4GB Recommended
Storage Space	500 MB

SECTION 7: PROJECT STATISTICS

Metric	Count
Python Files	12+
GUI Pages	14
DB Collections	8
MongoDB Pipelines	12+
Indexes Managed	15+
Lines of Code	3500+

Future Enhancements

- User auth & RBAC
- Email notifications for low stock
- PDF/Excel report generation
- FastAPI/Flask API backend
- Mobile companion app
- Real-time inventory sync
- Advanced analytics & forecasting
- Barcode scanning integration

SECTION 8: TROUBLESHOOTING

Common Errors & Solutions

Error / Symptom	Resolution
Cannot connect to MongoDB	Verify mongod service is running on localhost:27017.
Database not initialized	Run setup_database.py to seed the database.
No module named pymongo	Execute: pip install pymongo
Tkinter not found	Install standard tkinter package (e.g., sudo apt install python3-tk).
Slow performance	Check Index Management page; verify indexes exist and archive old data.
